1 Problem 1

In order to test that the vectors produced by my code were actually eigenvectors of $\mathbf{A}$, I verified that for $\epsilon' \approx \epsilon$ given as a parameter

$$\sum_i \left((\mathbf{A}v)_i - \lambda v_i\right)^2 < \epsilon'$$

The following bash commands are an overview of how to reproduce the results that follow.

```
# To build the programs, simply run make:

.../hw5$  make

# All three programs have usage statements. Examples are as follows

.../hw5$  ./eigP -e 1e-4 -k8 -f sample_eig.dat

# will find the 8 evecs with largest evals from input file sample.dat
# with error   1e-4.  eigenvalues and eigenvectors are written to
# respective *.out files

.../hw5$  ./eigP -e 1e-4 -k8 -r 100

# the same as above but with a random matrix of size N = 100 //$
```

Figure 2 demonstrates the pattern of convergence given varialbes ϵ and N for matrix $\mathbf{A}$. As one might expect, the number of iterations required for the power method to converge is proportional to the selected precision. However it is also interesting to note that the number of iterations to convergence *decreases* as the matrix size increases.

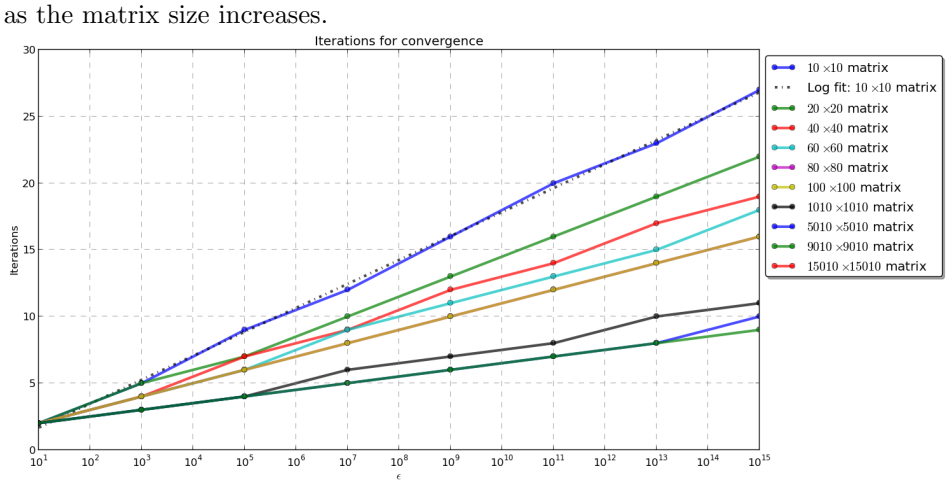

Figure 1: Plot of convergence iterations vs ϵ for selected matrix sizes

It can be noted that the number of iterations grew logarithmically with ϵ, i.e. for the 10×10 matrix,

$$\text{Iterations} \approx 0.780696 \times \log(\epsilon) - 0.130952$$

with an $r^2 = .9989$. Furthermore, as a function of matrix size for a fized ϵ, the number of iteration necessary as a function of matrix size N was close to the following,

$$\text{Iterations} \approx 49.5 \times N^{-.43} + 9$$

Therefore, we can say that as a function of ϵ, convergence is on the order of $\mathcal{O}(\log(\epsilon))$.

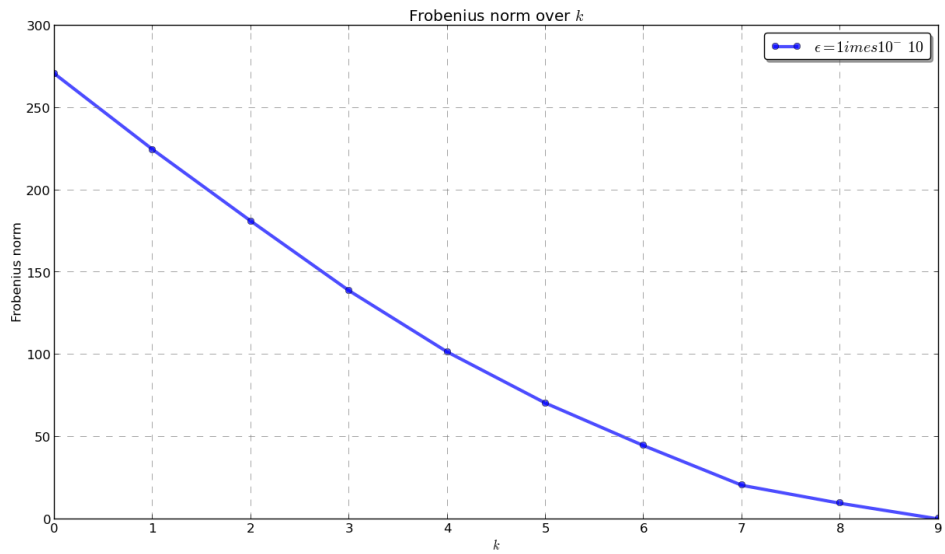

Figure 2: Plot of Frobenius norm vs k precision level $\epsilon = 1 \times 10^{-10}$